FIRST SEMESTER B.VOC(HONOURS) EXAMINATION

SCIENCE OF SOUND

PART A – Very Short Questions (2 Marks Each)

1. State one factor that affects the refraction of sound.

Variation in temperature/Change in density between 2 different mediums etc are major factors affecting refraction of sound.

2. State the function of a VU meter in one line.

A VU meter measures the average signal level of an audio signal.

3. Give a mixing action based on equal loudness curves (one line).

Boost bass and treble at low listening levels to maintain perceived balance.

4. State one condition required to perceive binaural beats

Two slightly different frequencies must be presented separately to each ear using headphones.

5. Name two auditory illusions.

Shepard Tone

McGurk Effect

6. Given a short attack and long release in an ADSR;name an instrument whose envelope this may resemble.

A violin or pad synthesizer shows short attack and long release.

7. Describe a short procedure to use pink noise for room level matching.

Play pink noise through speakers and adjust levels until equal SPL is measured in all locations using an SPL meter.

8. What is a critical band briefly?

A critical band is the frequency bandwidth of the ear within which sounds mask each other.

9. How does reverberation affect perceived distance of a sound?

More reverberation makes a sound appear farther away.

10. Define dynamic range.

Dynamic range is the difference between the loudest and softest sound levels in a system.

11. Explain briefly why sound is treated as a pressure wave in air.

Sound travels as alternating compressions and rarefactions of air pressure.

12. What SPL value is commonly cited as the threshold of pain?

The threshold of pain is approximately 120 dB SPL.

PART B – Short Questions (5 Marks Each)

13. Explain occupational exposure standards for sound (briefly cite typical dB/time limits) and their rationale.

Safe limit: 85 dB for 8 hours

Every 3 dB increase halves safe exposure time

Purpose: hearing protection

Prevents NIHL (Noise-Induced Hearing Loss)

14. Explain ITD and ILD cues and how they jointly provide horizontal localization Information.

ITD: Time difference between ears

ILD: Level difference between ears

Used for horizontal localization

Works mainly for mid–high frequencies

Brain combines both cues .

15. Explain how masking and noise floor affect useful dynamic range in recordings.

Noise floor hides low-level signals

Masking occurs when loud sound hides weaker sound

Reduces usable dynamic range

Affects clarity in recordings

Controlled by gain staging

16. Compare the harmonic content of sine, square, triangle and sawtooth waves, and explain why they sound different.

Sine: Only fundamental

Square: Odd harmonics

Triangle: Weak odd harmonics

Sawtooth: All harmonics

Timbre difference due to harmonics .

17. Explain how alignment level and operating level differ and why both matter.

Alignment: Reference calibration level

Operating level: Actual working level

Prevents distortion

Maintains signal-to-noise ratio

Ensures system compatibility .

18. Propose a brief instruction to teach society about preserving long-term hearing health in everyday life.

Avoid loud volumes

Use ear protection

Follow 60/60 rule

Take listening breaks

Regular hearing checkups .

19. Describe the Doppler effect and derive the qualitative pitch change when a source moves toward a listener.

Change in frequency due to motion

Approaching source: higher pitch

Receding source: lower pitch

Used in sirens & radar

Based on relative velocity .

20. Describe the functions of the outer, middle and inner ear in the chain of acoustic transduction.

Outer: Collects sound

Middle: Amplifies vibrations

Inner: Converts vibrations to nerve impulses

Ear ossicles named

Cochlea function included .

21. Define partials and overtones and describe how they shape perceived tone color.

Partials: All frequency components

Overtones: Above fundamental

Shape tone color

Define timbre

Instrument identification .

PART C – Essay Questions (10 Marks Each)

22. Explain Diffusion and reverberation. Compare its effects on acoustics.

Definition of diffusion.

Definition of reverberation

Differences

Acoustic effects

Studio & hall examples

Diagrams

Applications

Perceptual impact

RT60 mention

Summary

23. Describe dBu, dBV, dBm, LUFS metering.

dBu reference (0.775 V)

dBV reference (1 V)

dBm reference (1 mW)

LUFS loudness unit

Broadcasting use

Streaming standards

Audio metering importance

Peak vs loudness

Comparative table

Practical relevance

24. Explain the working principle of Cochlea in depth.

Structure

Organ of Corti

Hair cells

Basilar membrane

Frequency tuning

Mechanical–electrical conversion

Nerve impulse generation

Place theory

Traveling wave